

System Design (Telusko) - Day 8

One-Shot 5h course | Video 1:45-2:00

Sharding / Partitioning

Split a large dataset across multiple databases.

Key concepts

- Sharding: horizontal partitioning of data by a shard key.
- Strategies: range, hash, geo, directory-based.
- Hotspots happen with a bad shard key.
- Cross-shard queries and joins are hard - plan the key carefully.

Key example

shard = $\text{hash}(\text{user_id}) \% N$; keep related data on the same shard.

Today's tasks (1.5h)

- ☐ Pick a shard key for a chat app
- ☐ Explain a hotspot
- ☐ Note cross-shard query pain

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 1:45-2:00)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)